

Glasses: Enabling Fast Environment-aware Few-Shot Learning via Device-Cloud Collaboration

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Abstract

Pre-trained visual models have been deployed on various edge devices to facilitate a broad range of downstream tasks through few-shot learning (FSL), in particular when downstream data or on-device resources are limited. However, FSL often suffers from poor performance due to its inability to adapt to the characteristics of the deployment environments, while backbone fine-tuning prior to model deployment is typically infeasible because of the unavailability of environment-specific samples. To tackle this challenge, this paper presents Glasses, a lightweight fine-tuning scheme that can adapt ViT-based model backbones to deployment environments rapidly through device-cloud collaboration, helping the model achieve better FSL performance on the device. Glasses leverages the computational power and sample resources in the cloud to produce model updates for rapid model adaptation based on only one environment image without needing a label. Experiments with five models on two datasets demonstrate that Glasses can adapt a model rapidly and outperform the original backbone by 5.54%–22.56% in the 1-shot setting and by 2.50%–10.28% in the 5-shot setting. The source code is available at <https://github.com/CGCL-codes/Glasses>.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Computer systems organization** → **Cloud computing**; • **Human-centered computing** → **Ubiquitous and mobile computing**.

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Device-cloud collaboration, few-shot learning, environment-aware

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1 Introduction

As visual models are deployed on tremendous edge devices in diverse environments to facilitate AI applications like intelligent surveillance and manufacturing automation, few-shot learning (FSL) [10, 39] has emerged as a promising approach for adapting models to downstream tasks with minimal support data. However, they struggle to ensure the performance of a model when the deployment environment differs significantly from its pre-training conditions [26].

There has been substantial research on this *environment gap*. The key is to effectively and efficiently adapt the model to the deployment environment, e.g., lightning variations, background differences, etc. Some studies have revealed that irrelevant environmental information during image recognition can negatively affect model performance. Xiao et al. [43] point out that during pre-training, a model may mistakenly classify objects according to their surrounding environments rather than their own features. Another study [2] investigates how the performance of a model degrades when it is deployed in new environments. This challenge becomes particularly critical when a pre-trained model is deployed to fit a downstream task in a specific environment. The environment gap is further widened in FSL scenarios, where it is difficult to find sufficient downstream data to fit the model to the environment.

A series of studies have attempted to tackle this challenge. For example, COSOC [26] extracts foreground objects through image cropping to construct more discriminative features for classification.

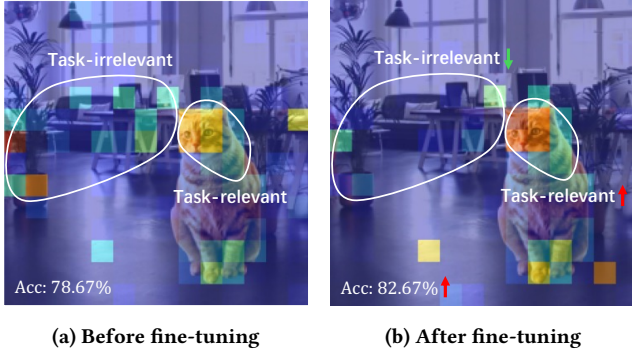


Figure 1: Visualization of attention distribution under consistent environmental conditions. (a) Before fine-tuning, the attention is dispersed across both the target object (the cat) and various background objects; and (b) after fine-tuning, the attention is focused more on the cat and much less on background objects.

FewTURE [15] and CPEA [13] focus on matching attention mechanisms to emphasize object-related information in images. However, these studies predominantly focus on enhancing the utilization of features extracted by the backbone. They often lead to suboptimal model performance and introduce additional inference overhead, which is particularly undesirable for resource-constrained edge devices.

To investigate the performance degradation, we conducted a 5-way 5-shot few-shot classification experiment using images captured under consistent environmental conditions. The resulting accuracy was only 78.67%, far from ideal. When examining the corresponding attention maps (see Figure 1a), we observed that the model’s attention was distracted by background objects like computers and potted plants. We hypothesize that this distraction caused by task-irrelevant environmental information plays a key role in model performance degradation. To validate this hypothesis, we synthesized 100 images with different foreground objects in a specific environment, and fine-tuned the backbone for 10 steps. The accuracy increased significantly by 4.00% to 82.67%, and the attention maps (see Figure 1b) showed a marked reduction in attention toward background objects. However, in practice, collecting labeled data is non-trivial, and such fine-tuning is a luxury for resource-constrained edge devices.

This paper presents Glasses, a lightweight fine-tuning scheme specifically designed to help ViT-based model backbones [9, 24, 37] learn to ignore the influence of task-irrelevant environmental information and focus on the informative ones, all with just a single unlabeled *environment image*, i.e., a photo of the scene without foreground objects (e.g., $\mathbf{x}_e$ in Figure 2). Glasses enhances model performance in FSL scenarios by rapid model adaptation to deployment environments via device-cloud collaboration.

Given an environment image from an edge device, Glasses employs a novel token-level environment infusion technique to integrate the environmental information from that image into the samples for model fine-tuning. Next, it employs multi-task auxiliary infusion-aware contrastive learning to guide the model to learn environmental knowledge effectively during fine-tuning. By

loading the fine-tuning parameters produced, usually a few MB, the pre-trained model on the edge device can significantly improve its downstream performance with zero additional inference overhead. Glasses is computation- and communication-efficient. For example, it transmits only around 3MB of data up and down and takes 27.33 seconds to adapt a ViT-T/16 model.

Our main contributions are as follows:

- We explore, from a new perspective, how environmental information affects the performance of a visual model when it is on an edge device via FSL.
- We propose Glasses, a novel scheme that leverages environmental information collected by the edge device and pre-training data in the cloud to fine-tune the model backbone. This process effectively teaches the model to focus on foreground targets and ignore irrelevant background objects in the deployment environment.
- Extensive experiments on both a synthetic dataset and a real-world dataset demonstrate that Glasses can effectively and efficiently adapt visual models to diverse environments.

2 Method

This section introduces Glasses in detail, starting with an overview (Section 2.1), followed by its two key techniques, i.e., token-level environment infusion (Section 2.2) and multi-task auxiliary infusion-aware contrastive learning (Section 2.3). At the end of this section, we briefly discuss extending token-level environment infusion to multi-image upload (Section 2.4).

2.1 Scheme Overview

Few-shot learning (FSL) involves a base dataset $\mathcal{D}_{\text{base}}$ and a novel dataset $\mathcal{D}_{\text{novel}}$, which contain disjoint classes. $\mathcal{D}_{\text{base}}$ is typically used to pre-train a backbone $f_{\theta}(\cdot)$ in the cloud. After pre-training, a set of N -way K -shot classification tasks $\mathcal{T} = \{(S_{\tau}, Q_{\tau})\}_{\tau=1}^{N_T}$ is constructed on $\mathcal{D}_{\text{novel}}$, where each task τ is built by sampling N novel classes and selecting K labeled samples per class to form the support set $S_{\tau} = \{(\mathbf{x}_{k,n}^{(\tau)}, y_{k,n}^{(\tau)})\}_{k,n=1}^{K,N}$, along with M query samples per class in the query set $Q_{\tau} = \{\mathbf{x}_{m,n}^{(\tau)}\}_{m,n=1}^{M,N}$. In practice, a few-shot model is often deployed in an environment where the background does not change significantly. When this model is distracted by task-irrelevant background objects, unique foreground object features are often submerged in the background object features. It will struggle to distinguish different foreground objects.

Glasses tackles this challenge with the support of the cloud. As shown in Figure 2, the edge device first captures an environment image $\mathbf{x}_e$ and uploads it to the cloud. The cloud initializes a model with the same pre-trained backbone as the on-device model (since the pre-trained backbone on the edge device is typically downloaded from the cloud, we assume an identical copy exists in the cloud by default), and fine-tunes it with LoRA [18] to adapt to the specific characteristics of the deployment environment. The adaptation modules (i.e., LoRA modules) are then transmitted back to the edge device to adjust the on-device backbone. During cloud-side fine-tuning, Glasses employs token-level environment infusion to augment the base dataset $\mathcal{D}_{\text{base}}$, and a novel infusion-aware contrastive learning mechanism with multi-task auxiliary objectives to

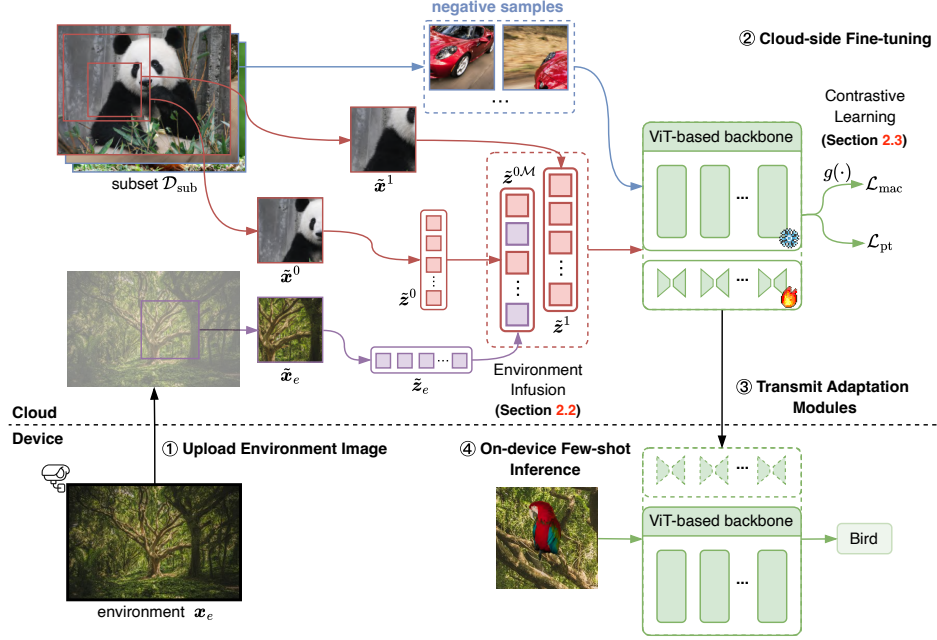


Figure 2: Glasses involves four steps. ① The edge device captures an environment image x_e and uploads it to the cloud. ② The cloud server randomly selects samples from the cloud dataset $\mathcal{D}_{\text{base}}$ and fine-tunes the model backbone based on the environment image x_e . ③ The cloud server transmits backbone adaptation modules to the edge device. ④ The edge device performs few-shot inference with the received adaptation modules.

fine-tune the backbone based on the augmented dataset. The fine-tuned backbone will be able to ignore task-irrelevant environmental information and focus on foreground objects.

2.2 Token-level Environment Infusion

As discussed above, the backbone needs to be fine-tuned to adapt to the deployment environment. However, fine-tuning requires data, and insufficient data often leads to suboptimal model performance. Obtaining sufficient fine-tuning data in the deployment environment is often difficult, if not impossible, especially in the context of FSL where supervision is limited by design. In the meantime, gathering extra, environment-aligned samples on the device side is often infeasible. To address this challenge, Glasses employs a novel environment infusion technique to leverage rich data from the cloud. Specifically, it incorporates environmental information into the samples retrieved from the pre-trained dataset $\mathcal{D}_{\text{base}}$ for backbone fine-tuning.

Environment Tokenization. When training begins in the cloud under the Glasses scheme, a subset $\mathcal{D}_{\text{sub}}$ is randomly sampled from the cloud’s pre-training dataset $\mathcal{D}_{\text{base}}$, which was originally used to pre-train the visual feature extractor $f_\theta(\cdot)$. For each sample $x \in \mathcal{D}_{\text{sub}}$, it constructs two distinct augmented views, $\tilde{x}^0$ and $\tilde{x}^1$. Then, its patch-token representation is obtained through

$$z = \phi(x) \in \mathbb{R}^{T \times d}, \quad (1)$$

where $\phi(\cdot)$ denotes the ViT patch embedding function, T is the number of tokens (including the class token), and d is the hidden dimension. Two tokens $\tilde{z}^0$ and $\tilde{z}^1$, are then constructed from $\tilde{x}^0$ and

$\tilde{x}^1$, for environment infusion and contrastive training. In each fine-tuning batch, Glasses randomly crops a region from the uploaded environment image x_e , whose patch embedding is denoted as $\tilde{z}_e$. This serves as the source of environment tokens for constructing mixed views, allowing Glasses to inject environmental context into synthetic training samples.

Infusion Masking. Glasses constructs a T -bit binary mask $\mathcal{M} \in \{0, 1\}^T$ over the token sequence indexed by $t=0, \dots, T-1$, where $t=0$ corresponds to the [CLS] token. The mask is initialized as

$$\mathcal{M}_0 = 0, \quad \mathcal{M}_t \sim \text{Bernoulli}(p) \text{ for } t = 1, \dots, T-1, \quad (2)$$

where p is the infusion ratio. In other words, all patch tokens (i.e., all tokens except [CLS]) are independently replaced by their environment counterparts with probability p , which realizes token-level environment infusion. The infused view is then obtained by token-wise selection:

$$\tilde{z}^{0M} = (1 - \mathcal{M}) \odot \tilde{z}^0 + \mathcal{M} \odot \tilde{z}_e. \quad (3)$$

Glasses feeds both the infused samples $\tilde{z}^{0M}$ and their clean counterparts $\tilde{z}^1$ into the backbone during fine-tuning.

Theoretical Justification. In deployments where the uploaded environment image is predominantly background, token-level environment infusion is most effective at moderate replacement ratios (e.g., $p \approx 0.5$). It strikes a balance between aligning background statistics and preserving foreground semantics. Intuitively, injecting environment tokens lets the backbone encounter target-background features during training. It narrows the gap between the pre-training distribution $\mathcal{D}_{\text{base}}$ and the deployment distribution $\mathcal{D}_{\text{novel}}$, without overwhelming the instance with background-only content.

In expectation, token-level environment infusion is equivalent to empirical risk minimization on a token-level mixture distribution Q_p :

$$\mathbb{E}_{\mathcal{M}}[\ell(f_{\theta}(\tilde{z}^{0M}), y)] = \mathbb{E}_{(z^*, y) \sim Q_p}[\ell(f_{\theta}(z^*), y)], \quad (4)$$

where each token is assigned from the environment with probability p and from the base sample with probability $1 - p$. This view shows that without extra labels, the backbone is optimized directly on an “environment-aligned” training distribution.

Moreover, if backbone f_{θ} and loss ℓ are Lipschitz, there exists a constant L such that

$$\left| \mathbb{E}_{\mathcal{M}}[\ell(f_{\theta}(\tilde{z}^{0M}), y)] - \ell(f_{\theta}(\tilde{z}^0), y) \right| \leq L \sqrt{p} \|\tilde{z}_e - \tilde{z}^0\|_2. \quad (5)$$

As p increases, the model sees more background information from the target environment, but the difference from the original (uninfused) sample is still kept within a small bound. Replacing some foreground tokens also works in a way similar to using a regularization strategy—similar to dropout or label smoothing—which helps the model become more robust against background changes.

Overall, choosing a moderate p provides theoretical guarantees for environment alignment without sacrificing discriminative semantics, and empirically yields a favorable trade-off between adaptation and accuracy. A formal analysis, including mixture-risk equivalence and deviation bounds, is provided in Appendix A.

Through this mechanism, Glasses leverages the tokenized patch structure of ViT to encourage the backbone to develop environment-aware attention during fine-tuning. In this way, it suppresses task-irrelevant background cues while extracting discriminative features for classification.

2.3 Multi-task Auxiliary Infusion-aware Contrastive Learning

Optimization Objective and Justification. As discussed in Section 1 and illustrated in Figure 1, the backbone’s sensitivity to task-irrelevant environmental information compromises the performance of FSL. Model adaptation must reduce the influence of environmental information on the backbone’s feature extraction process. To achieve this goal, we design a multi-task auxiliary and infusion-aware contrastive loss for model fine-tuning. It aims to enable personalized model adaptation to the environment while preserving the backbone’s original feature extraction capability.

The optimization objective is to maintain stable performance on the original pre-training tasks while reducing the impact of environmental information introduced into the training samples by the token-level environment infusion method discussed in Section 2.2 on feature extraction.

Infusion-Aware Contrastive Loss. Inspired by previous contrastive learning algorithms [6, 14, 28], Glasses employs a novel infusion-aware contrastive loss tailored to guide the backbone in disregarding task-irrelevant environmental information. Specifically, it maximizes the agreement between two differently views of the same sample, $\tilde{z}_i^{0M}$ and $\tilde{z}_i^1$, while minimizing the agreement between features of different samples infused with environmental information, such as $\tilde{z}_i^{0M}$ and $\tilde{z}_j^{0M}$, in the latent space.

To obtain representations in the latent space, each input is first passed through the feature extractor $f_{\theta}(\cdot)$, followed by a projection

head $g(\cdot)$ that maps the features into a space where the contrastive loss is applied:

$$\mathbf{y}_i^{0M} = g(f_{\theta}(\tilde{z}_i^{0M})), \quad \mathbf{y}_i^1 = g(f_{\theta}(\tilde{z}_i^1)), \quad (6)$$

where $g(\cdot)$ is implemented as a single-layer fully connected network.

Given the latent representations $\mathbf{z}_i^{0M}$ and $\mathbf{z}_i^1$ obtained from two different views of the same sample, the infusion-aware contrastive loss for an example $\mathbf{x}_i$ is defined as:

$$\ell_i = -\log \frac{\exp(\text{sim}(\mathbf{y}_i^{0M}, \mathbf{y}_i^1) / \tau_{pos})}{\sum_{k=1}^N \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{y}_i^{0M}, \mathbf{y}_k^{0M}) / \tau_{neg})}, \quad (7)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, $\mathbb{1}_{[k \neq i]} \in \{0, 1\}$ is an indicator function that returns 1 iff $k \neq i$, and τ_{pos} and τ_{neg} are temperature parameters for positive and negative pairs, respectively.

Glasses uses separate temperature parameters τ_{pos} and τ_{neg} to control the sharpness of similarities for positive and negative pairs. For positive pairs, where two views of the same sample differ by whether environmental information is infused, Glasses employs a lower τ_{pos} to encourage tighter feature closeness. This encourages the backbone to focus on task-relevant features and ignore irrelevant information introduced by the environment. For negative pairs, which consist of different samples infused with environmental information, a higher τ_{neg} is used to maintain a clear separation between samples.

The overall infusion-aware contrastive loss $\mathcal{L}_{mac}$ over a mini-batch of size N is computed by averaging the per-sample losses:

$$\mathcal{L}_{mac} = \frac{1}{N} \sum_{k=1}^N \ell_k. \quad (8)$$

With infusion-aware contrastive loss, Glasses can effectively adjust the relative distances between different types of samples in the feature space. This promotes the alignment between representations of samples with infused environmental information and those without. It enables the backbone to better adapt to the deployment environment by suppressing task-irrelevant environmental features.

Multi-task Auxiliary Learning. Multi-task learning [5] enables a model to simultaneously learn multiple tasks to capture more general and discriminative features. To preserve the feature extraction ability of the backbone and prevent overfitting to environmental information, Glasses incorporates the tasks pre-trained for the backbone within its contrastive learning process.

For different backbone pre-training tasks, Glasses employs task-specific loss functions $\mathcal{L}_{pt}$, such as cross-entropy for supervised tasks, and DINO [4], iBOT [49], and other losses for self-supervised tasks. These are formally defined as follows:

$$\mathcal{L}_{pt} = \begin{cases} \mathcal{L}_{CE} & \text{for classification pre-training} \\ \mathcal{L}_{DINO} & \text{for DINO pre-training} \\ \mathcal{L}_{iBOT} & \text{for iBOT pre-training} \\ \vdots & \text{for other pre-training tasks} \end{cases} \quad (9)$$

Table 1: Few-shot classification accuracy (%) for 5-way 1-shot and 5-way 5-shot on Focura and Street using different ViT-based backbones. ✓ / ✗ denote with/without Glasses. Averages with 95% confidence intervals (CIs) are reported; for Street, we show mean ($\pm$ CI) per the evaluation protocol, while for Focura (10 scenes), we report averages here only. See Appendix D.1 for more details of the results.

Backbone	Metric	Focura				Street			
		1-shot		5-shot		1-shot		5-shot	
		✗	✓	✗	✓	✗	✓	✗	✓
DeiT-T/16	$\ \cdot\ ^2$ cos	59.24	73.75	82.56	88.02	54.21 $\pm$ 1.43	62.31 $\pm$ 1.44	73.43 $\pm$ 0.97	79.20 $\pm$ 0.99
		61.83	75.27	80.97	87.14	54.11 $\pm$ 1.45	61.55 $\pm$ 1.46	73.44 $\pm$ 0.99	78.59 $\pm$ 0.96
DeiT-S/16	$\ \cdot\ ^2$ cos	58.85	81.08	88.28	93.95	55.47 $\pm$ 1.61	66.27 $\pm$ 1.57	74.85 $\pm$ 1.13	81.60 $\pm$ 1.07
		66.70	84.20	87.14	93.60	57.89 $\pm$ 1.72	67.24 $\pm$ 1.50	75.67 $\pm$ 1.17	81.87 $\pm$ 1.06
ViT-B/16	$\ \cdot\ ^2$ cos	71.87	81.22	88.25	92.77	45.25 $\pm$ 1.67	62.17 $\pm$ 1.62	70.59 $\pm$ 1.21	80.87 $\pm$ 1.04
		71.85	83.60	87.72	92.33	45.59 $\pm$ 1.60	63.25 $\pm$ 1.56	70.89 $\pm$ 1.23	80.51 $\pm$ 1.06
Swin-T/7	$\ \cdot\ ^2$ cos	64.37	82.26	89.63	94.56	52.89 $\pm$ 1.34	59.12 $\pm$ 1.51	71.07 $\pm$ 1.19	75.32 $\pm$ 1.04
		69.85	85.09	89.01	94.13	55.05 $\pm$ 1.49	60.59 $\pm$ 1.50	72.45 $\pm$ 1.24	75.64 $\pm$ 1.09
Swin-S/7	$\ \cdot\ ^2$ cos	62.43	84.99	88.35	95.18	50.27 $\pm$ 1.46	61.80 $\pm$ 1.48	71.24 $\pm$ 1.29	79.60 $\pm$ 1.15
		69.46	87.12	88.69	94.86	53.95 $\pm$ 1.45	62.19 $\pm$ 1.43	73.65 $\pm$ 1.30	79.64 $\pm$ 1.15

Thus, the complete multi-task auxiliary infusion-aware contrastive loss can be expressed as:

$$\mathcal{L}_{all} = \mathcal{L}_{mac} + \mathcal{L}_{pt}. \quad (10)$$

2.4 Extension to Multi-image Upload

Glasses can also accommodate multiple environment images captured at different times or from different viewpoints. On the cloud, it constructs an environment-token pool $\mathcal{P}_e$ by tokenizing all uploaded images. During token-level environment infusion (Section 2.2), it samples mask $\mathcal{M}$ as usual, while drawing replacement tokens from $\mathcal{P}_e$ (optionally with scene- or view-aware stratified sampling to preserve diversity). This variant keeps labels unchanged and reuses the objectives presented in Section 2.3. The difference is their reliance on different sources of environment tokens. In this way, Glasses can adapt to various environmental factors, e.g., lighting, viewpoint, clutter, without incurring inference overhead. The ability of Glasses to utilize multiple images is evaluated in Section 3.3.

3 Experiments

3.1 Experimental Setup

Datasets. Due to the lack of public image datasets that contain different foreground objects with a consistent background, we built two datasets for the evaluation of Glasses. Inspired by [46], we synthesized a dataset called *Focura* that contains images with different foreground objects in the same background. This dataset contains 10 scenes, 10 categories, and 20 images per category. In addition, we collected photos taken from a real-world street to construct a *Street* dataset. This dataset contains images of different foreground objects across 5 categories captured in a fixed environment, with at least 20 images in each category. For fair comparison, all input images are resized to 224×224 . More details about the datasets are provided in Appendix B.

Backbones. To evaluate the performance of Glasses comprehensively, we conduct experiments with a variety of publicly available ViT-based backbones, including different variants of ViT [9], DeiT [37], and Swin Transformer [24], as well as ViT backbones pre-trained on different tasks [4, 49]. To facilitate a fair comparison, all baselines use the same pre-trained backbones as Glasses. In this way, the evaluation avoids performance differences caused by variations in backbone initialization.

Implementation Details. Environment-aware fine-tuning is performed with an NVIDIA A40 GPU (48 GB). ImageNet-1K serves as the cloud dataset $\mathcal{D}_{base}$, which is used to pre-train the backbone network. From $\mathcal{D}_{base}$, we randomly sample 5,120 images to form a training subset $\mathcal{D}_{sub}$ for the backbone fine-tuning process in Glasses. Fine-tuning is conducted for a single epoch with a batch size of 128, using the AdamW [25] optimizer and a learning rate of 0.0008. The infusion ratio p is set to 0.5, as suggested by the theoretical analysis in Appendix A and the sensitivity analysis in Appendix C.

Evaluation Protocol. The performance of Glasses is evaluated on 5-way 1-shot and 5-way 5-shot classification tasks. The average accuracy over 100 episodes randomly sampled from the evaluation set, each with 15 query images per class, is reported along with the 95% confidence intervals following the standard evaluation protocol. Unless otherwise specified, we adopt ProtoNet [34], a widely adopted benchmark in studies of few-shot learning [23, 40, 41], as the default few-shot inference method in step ④ of Glasses.

3.2 Evaluation

The performance of Glasses is evaluated from four perspectives.

Performance across Backbones. We run Glasses to adapt five different ViT-based backbones, measuring the distance between sample features and class prototypes for classification with two common similarity metrics, i.e., cosine similarity and Euclidean distance. Table 1 summarizes the results. It can be observed that Glasses can improve the accuracy of these models by 2.50%–10.28%

Table 2: Classification accuracy comparison on the Focura and Street datasets under the 5-way 1-shot and 5-way 5-shot settings. Here, '  | [Method] ' denotes the use of Glasses combined with the few-shot learning method [Method]. See Appendix D.2 for more details of the results.

Model	Backbone	Pre-training Task	Focura		Street	
			1-shot	5-shot	1-shot	5-shot
ProtoNet _{ℓ₂} [34]	ViT-S/16	classification	58.85	88.28	55.47±1.61	74.85±1.13
ProtoNet _{cos}	ViT-S/16	classification	66.70	87.14	57.89±1.72	75.67±1.17
FewTURE [15]	ViT-S/16	iBOT	45.36	58.60	38.85±0.99	59.04±1.27
CPEA [13]	ViT-S/16	iBOT	73.28	89.82	67.60±1.27	78.36±1.46
BPA [33]	ViT-S/16	iBOT	80.04	90.84	60.82±1.49	79.85±1.20
 ProtoNet _{cos}	ViT-S/16	classification	84.20	93.60	67.24±1.50	81.87±1.06
 ProtoNet _{cos}	ViT-S/16	iBOT	84.38	93.58	68.61±1.48	82.35±1.02

Table 3: Classification accuracy comparison on the Focura and Street datasets under the 5-way 0-shot settings.

Model	Backbone	Focura	Street
CLIP [30]	ViT-B/32	93.14	61.25±0.49
 CLIP	ViT-B/32	95.56	71.68±0.42
SigLIP [45]	ViT-B/32	97.95	67.87±0.32
 SigLIP	ViT-B/32	98.37	71.68±0.42
SigLIP 2 [38]	ViT-B/32	98.03	71.71±0.40
 SigLIP 2	ViT-B/32	98.73	73.52±0.46

under the 5-shot setting. Notably, under the 1-shot setting, Glasses achieves even more substantial performance gains, with an average accuracy improvement of 13.20%, and the largest improvement reaching 22.56% (from 62.43% to 84.99%).

Comparison against Few-shot Baselines. Table 2 compares Glasses against five baselines, all evaluated using a ViT-S/16 backbone pre-trained or fine-tuned on ImageNet-1k. It is evident that Glasses outperforms all baselines. In 5-shot scenarios, Glasses achieves an improvement of up to 2.76% over the best-performing baseline, while in the 1-shot scenario, the improvement is even more significant, reaching 7.19%. Notably, even with a traditionally supervised pre-trained ViT as the backbone, which is generally considered unsuitable for few-shot fine-tuning, Glasses still surpasses all baselines, achieving accuracy comparable to models based on self-supervised pre-trained ViTs.

Furthermore, it is worth noting that FewTURE [15] exhibits unusually low accuracy, even worse than ProtoNet, the simple baseline ProtoNet [34]. The reason is that FewTURE relies on token-level comparisons between the query and support sets. When a large number of tokens conveying similar environmental information are present as noise, a situation rarely encountered in standard datasets such as MiniImageNet [39], FewTURE struggles to distinguish foreground objects submerged in background objects effectively. This highlights the importance of Glasses to focus a model on foreground objects and ignore task-irrelevant environmental information.

Impact on Zero-shot Learning. Table 3 reports the performance of Glasses when it is integrated with typical zero-shot vision-language models, including CLIP [30], SigLIP [45], and SigLIP 2 [38], which represents the state-of-the-art in zero-shot vision-language

Table 4: Glasses performance on ViT-S/16 across different pre-training tasks.

Pre-training Task	Glasses	Focura	
		1-shot	5-shot
Classification	✗	66.70	87.14
	✓	84.20	93.60
DINO [4]	✗	70.19	88.06
	✓	82.61	92.71
iBOT [49]	✗	72.38	89.48
	✓	84.38	93.58

models. We observe consistent gains across both datasets. On Focura, Glasses improves the accuracy of CLIP from 93.14% to 95.56%, and further enhances SigLIP 2 from 98.03% to 98.73%. On the more challenging Street dataset, the improvements are even more pronounced, evidenced by an improvement of +10.43% for CLIP and +1.81% for SigLIP 2. These results demonstrate that Glasses not only enables environment-aware adaptation offered by few-shot classifiers, but also strengthens the robustness and transferability of state-of-the-art zero-shot models. In particular, the ability to suppress spurious background correlations makes zero-shot recognition more reliable in unseen environments.

Impact of Pre-training Tasks. As shown in Table 4, Glasses consistently improves model performance across different pre-training paradigms, including supervised classification and self-supervised methods (DINO [4] and iBOT [49]). The results demonstrate that regardless of the pre-training loss $\mathcal{L}_{pt}$, Glasses enables the backbone to adapt to deployment environments effectively. Notably, it achieves average gains of 13.97% in the 1-shot setting and 5.07% in the 5-shot setting compared to baselines without Glasses, confirming its robustness across diverse pre-training tasks.

3.3 Performance on Multi-image Upload

We evaluate the accuracy of ViT-S/16 with and without the support of Glasses by progressively increasing the number of environment images uploaded from Focura and constructing the respective training subsets, as shown in Figure 3. When different numbers of scenes are uploaded, Glasses yields substantial accuracy improvements. It means 1-shot accuracy rises from 55.00% to 81.82%, while 5-shot accuracy increases from 78.29% to 92.92%. As the number

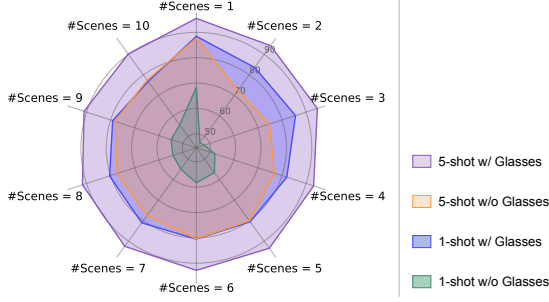


Figure 3: Glasses performance on ViT-S/16 across different numbers of uploaded scenes.

Table 5: Ablation study on Focura using ViT-S/16 as the evaluation backbone.

Env-Infusion	Multi-Task		Focura	
	$\mathcal{L}_{\text{mac}}$	$\mathcal{L}_{\text{pt}}$	1-shot	5-shot
w/o Fine-tuning			66.70	87.14
	✓		65.52	82.66
	✓	✓	76.16	90.94
✓	✓		76.45	87.69
✓		✓	77.87	92.01
✓	✓	✓	84.20	93.60

of uploaded scenes increases, the baseline degrades more rapidly: in the 5-shot setting, its accuracy falls from 87.66% with a single scene to 77.49% with ten scenes, a decline of 10.18%. In contrast, Glasses shows a much gentler decline, dropping only from 95.48% to 90.17% (a reduction of 5.31%) over the same range, while consistently maintaining higher accuracy. This robustness enables Glasses to adapt effectively to multi-scene environments—indeed, 1-shot with Glasses often matches or even surpasses the 5-shot baseline, thereby markedly reducing the reliance on labeled shots.

We further observe a sharp drop in the 1-shot accuracy of the baseline from 68.26% to 46.93% when the number of scenes increases from 1 to 2. An analysis of misclassified samples suggests that in the 1-shot case, the support set is too sparse. Its background is highly similar within the same scene, but differs substantially from a new scene. This causes prototype estimation to be biased by background cues, leading to significant classification errors. As the number of scenes increases further, the expanded scene diversity gradually dilutes this background-induced bias, and the baseline accuracy partially recovers. In contrast, with the support of Glasses, the model does not suffer from this issue. In the two-scene setting, it still attains an accuracy of 83.63% for 1-shot. This indicates that Glasses successfully suppresses task-irrelevant background interference during feature learning.

Overall, this experiment demonstrates the scalability and effectiveness of Glasses under multi-view/multi-scene settings.

3.4 Ablation Study

Table 5 presents the results of an ablation study conducted on Focura with ViT-S/16 as the backbone. Without fine-tuning, the model achieves an accuracy of 66.70% in the 1-shot setting and 87.14% in the 5-shot setting. With $\mathcal{L}_{\text{mac}}$ only, i.e., the infusion-aware contrastive loss, the model accuracy decreases by 1.18%–4.48%.

Table 6: Model adaptation time. Downlink = 20Mbps, Uplink = 10Mbps upload, Environment Image Size = 1.5MB.

Backbone	Training Time (s)	Traffic Volume (MB)	Overall Time (s)
ViT-T/16	25.47	1.63	27.33
ViT-S/16	39.00	3.22	41.49
ViT-B/16	51.71	6.38	55.47
Swin-T/7	49.06	3.09	51.49
Swin-S/7	71.34	6.30	75.06

This indicates that the contrastive signal alone does not suffice to preserve the model’s original feature extraction capability. When Glasses combines $\mathcal{L}_{\text{mac}}$ and the pre-training task loss ($\mathcal{L}_{\text{pt}}$), the model performance improves significantly. This highlights the complementary roles of contrastive supervision and pre-training objectives (Section 2.3), which enable the model to learn how to adapt to environmental information while maintaining generalizable representations. Even in the absence of contrastive learning, Glasses remains effective: using only token-level environment infusion (Section 2.2) and $\mathcal{L}_{\text{pt}}$, it improves accuracy by 11.17% in the 1-shot setting and 4.87% in the 5-shot setting.

Finally, integrating environment infusion, $\mathcal{L}_{\text{mac}}$, and $\mathcal{L}_{\text{pt}}$, Glasses achieves the best results, with an accuracy of 84.20% and 93.60% in 1-shot and 5-shot settings, respectively. These results validate the importance of environment infusion and multi-task auxiliary infusion-aware contrastive learning, as well as their joint effectiveness in improving few-shot model adaptation to the deployment environment.

3.5 Time Cost

Table 6 presents the overall time taken for Glasses to adapt various ViT-based models, calculated based on the time to fine-tune the backbone in the cloud and the sizes of fine-tuning parameters transmitted to the device. These results demonstrate the lightweight nature of Glasses, with the entire process taking as little as 27.33s, and a maximum of 75.06s. Based on GPU rental prices¹, adapting the most time-consuming backbone (Swin-S/7) would cost only \$0.0085. Glasses is also communication-efficient, transmitting only 1.63–6.30MB in total. This ensures its high efficiency even under poor network conditions.

3.6 Visualization Analysis

Figure 4a and Figure 4b present the attention heatmaps before and after backbone adaptation with Glasses. It can be observed that Glasses effectively suppresses the backbone’s attention to task-irrelevant environmental information and promotes a more focused extraction of task-relevant, non-environmental features. Figure 4c and Figure 4d present the t-SNE visualizations of the features extracted by the backbone before and after adaptation with Glasses. After adaptation, the model effectively suppresses similar environmental information across different samples, leading to more compact intra-class clustering and better inter-class separation in the feature space, which translates to an enhanced discriminative capability.

¹The rental price of NVIDIA A40 comes from <https://autodl.com/home>.

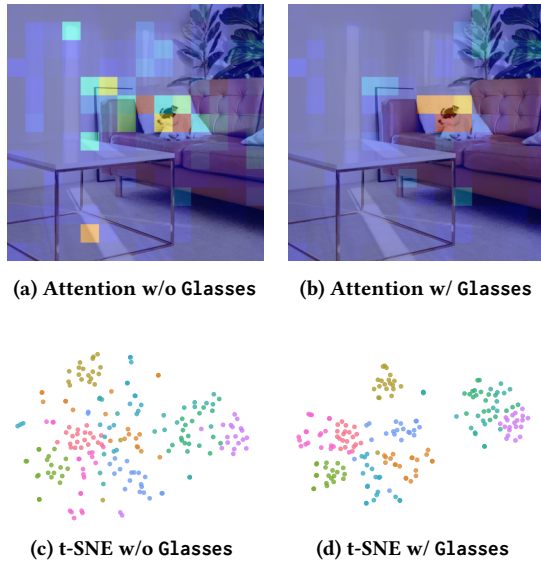


Figure 4: Visualization of the effect of Glasses. (a) Without Glasses, the attention is diffusely distributed across the dog and background objects; (b) With Glasses, the attention becomes concentrated on the dog; (c) Without Glasses, feature representations appear more scattered in the embedding space; (d) With Glasses, features from the same class form more compact clusters while maintaining better separation between different classes.

4 Related Work

Visual Model Deployment. Deploying pre-trained models on edge devices for downstream tasks has become a common practice in modern AI applications [29, 42]. However, the discrepancy between the objectives and the formats of pre-training and downstream tasks often leads to suboptimal performance and transfer difficulties, making model adaptation an imperative research topic [7, 44]. In the field of computer vision, popular model adaptation methods include fine-tuning [17, 18] on downstream datasets, prompt learning [20, 21, 50], and few-shot learning [19, 48]. Among these, few-shot learning is gaining significant attention due to its low data requirements and flexibility, which makes it an ideal deployment option for resource-constrained edge devices like smart cameras [32].

Few-shot Image Classification. Few-shot image classification methods can be broadly categorized into two main groups: optimization-based and metric-based. Optimization-based methods [11, 19, 27, 31] focus on training meta-learners that rapidly adapt to novel classes using only a few labeled examples and limited gradient steps. These methods often require on-device model fine-tuning at deployment, which is often a grand challenge for resource-constrained edge devices, sometimes a mission impossible. Moreover, they typically rely on specialized mechanisms during model pre-training, such as the meta-training mechanism used in MAML [11], a classic optimization-based method that initializes models for fast adaptation to downstream tasks. This reliance limits their generalizability in real-world scenarios that leverage pre-trained models.

In contrast, metric-based methods aim to learn a transferable feature space in which classification is performed by comparing query samples to class prototypes or exemplars based on appropriate similarity measures. Recent studies have attempted to improve model performance by enhancing backbone pre-training [8, 12, 35] and refining the design of similarity metrics [1, 22, 34, 47]. Among these, local feature-based methods [13, 15, 16, 26, 33] have excelled by focusing on task-relevant regions and reducing the influence of irrelevant background information.

Nonetheless, an important but often overlooked observation is that in many real-world scenarios, the deployment environment of the device tends to remain relatively stable. This environmental consistency offers a valuable opportunity: by leveraging such stability to adapt the backbone representation to the target environment, models can suppress the impact of environmental noise and enhance attention to task-relevant features. Such environment-aware adaptation has the potential to significantly improve the performance and robustness of few-shot classification models in practical applications.

Impact of Environment. The impact of the environment on the performance of visual models has been studied extensively. Torralba et al. [36] provide early evidence of how background cues influence visual predictions. Beery et al. [2] demonstrated that the performance of a model significantly degrades when models are deployed in unseen environments. They used camera traps to highlight this challenge and advocated for improved generalization across domains. In the few-shot setting, Luo et al. [26] showed that training with a mix of samples with and without background information, while evaluating on foreground-only inputs, leads to higher model accuracy. This finding motivated the token-level environment infusion strategy in Glasses, which helps the model better separate useful task-related cues from environmental noise, enabling it to extract more informative features for downstream decision-making.

5 Conclusion

This paper presented Glasses, a lightweight fine-tuning mechanism based on device-cloud collaboration that enables rapid adaptation of ViT-based model backbones to deployment environments. It is capable of improving the performance of a model in few-shot learning tasks by introducing the token-level environment infusion strategy, which incorporates environmental information into general datasets through data augmentation, and a multi-task auxiliary infusion-aware contrastive learning mechanism, which guides the backbone to focus on task-relevant object features. Extensive experiments demonstrate the state-of-the-art performance of Glasses in adapting various pre-trained visual models to their deployment environments. In the future, we will explore the potential of Glasses in adapting other Transformer-based visual models, such as DETR [3] and similar architectures.

6 Acknowledgments

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A Theoretical Analysis of Environment Infusion

This appendix formalizes token-level environment infusion. We show it equals ERM on a token-level mixture distribution and derive tight deviation bounds from training on either pure base or pure environment tokens. We also justify a moderate replacement ratio (e.g., $p \approx 0.5$) from a minimax perspective and give a high-probability guarantee for preserving foreground tokens.

A.1 Setup and Assumptions

Let $\tilde{z}^0 \in \mathbb{R}^{T \times d}$ be the tokenized base sample and $\tilde{z}_e$ the environment tokens extracted from the uploaded image. Let P be the number of patch tokens (excluding [CLS]), and infusion acts only on patch tokens (no infusion on [CLS]). Draw an i.i.d. mask $\mathcal{M} \in \{0, 1\}^P$ with $M_i \sim \text{Bernoulli}(p)$. Define $\Delta = \tilde{z}_e - \tilde{z}^0$.

Let $f_\theta : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^k$ be the backbone and $\ell : \mathbb{R}^k \times \mathcal{Y} \rightarrow \mathbb{R}_+$ the loss. Assume: (i) f_θ is L_f -Lipschitz w.r.t. its input; (ii) $\ell(\cdot, y)$ is L_ℓ -Lipschitz in its first argument. Let $L = L_f L_\ell$ and abbreviate

$$R_\theta(z; y) \triangleq \ell(f_\theta(z), y). \quad (11)$$

A.2 Infusion-Risk Equivalence

Define a token-level mixture distribution Q_p : at each token position, the source is drawn from the environment with prob. p and from the base with prob. $1 - p$.

LEMMA A.1 (ERM ON A MIXTURE DISTRIBUTION).

$$\mathbb{E}_{\mathcal{M}}[R_\theta(\tilde{z}^{0\mathcal{M}}; y)] = \mathbb{E}_{(z^*, y) \sim Q_p}[R_\theta(z^*; y)]. \quad (12)$$

PROOF SKETCH. Independent token-wise masking induces Q_p . Swapping the expectation over $\mathcal{M}$ with that over Q_p yields the claim. $\square$

A.3 Deviation Bounds from Base and Environment

The following bounds show smooth, controlled deviation from training on pure base or pure environment tokens.

PROPOSITION A.2 (DEVIATION FROM BASE).

$$|\mathbb{E}_{\mathcal{M}} R_\theta(\tilde{z}^{0\mathcal{M}}; y) - R_\theta(\tilde{z}^0; y)| \leq L \sqrt{p} \|\Delta\|_2. \quad (13)$$

PROPOSITION A.3 (DEVIATION FROM ENVIRONMENT).

$$|\mathbb{E}_{\mathcal{M}} R_\theta(\tilde{z}^{0\mathcal{M}}; y) - R_\theta(\tilde{z}_e; y)| \leq L \sqrt{1-p} \|\Delta\|_2. \quad (14)$$

PROOF SKETCH. By Lipschitzness of f_θ and ℓ ,

$$|R_\theta(\tilde{z}^{0\mathcal{M}}; y) - R_\theta(\tilde{z}^0; y)| \leq L \|\mathcal{M} \odot \Delta\|_2. \quad (15)$$

Taking expectation and using Jensen, $\mathbb{E} \|\mathcal{M} \odot \Delta\|_2 \leq \sqrt{\mathbb{E} \|\mathcal{M} \odot \Delta\|_2^2} = \sqrt{p} \|\Delta\|_2$. Replacing $\mathcal{M}$ with $(1 - \mathcal{M})$ gives (14). $\square$

A.4 Why a Moderate Ratio p ? A Minimax View

The term $\max\{\sqrt{p}, \sqrt{1-p}\}$ is minimized at $p = \frac{1}{2}$. Thus, from a minimax perspective that jointly controls deviation from both base and environment risks, $p^* = \frac{1}{2}$ is a natural choice. This matches the practical “background-only” setting: too large p risks overwhelming foreground semantics, whereas a moderate p aligns background while preserving discriminative cues.

A.5 Foreground Preservation with High Probability

Let $\mathcal{F} \subseteq \{1, \dots, P\}$ index F foreground tokens. Each is retained with prob. $1 - p$, so $R \sim \text{Binomial}(F, 1 - p)$ is the retained count. For any $\varepsilon > 0$, Hoeffding/Chernoff yields

$$\Pr(R \leq (1 - p - \varepsilon)F) \leq \exp(-2\varepsilon^2 F). \quad (16)$$

Hence, for moderate p and nontrivial F , a $(1 - p - \varepsilon)$ fraction of foreground tokens is preserved with high probability, preventing degenerate all-background cases.

A.6 Regularization Effect

Stochastic replacement induces bounded label-input mismatch, controlled by (13)–(14). In practice, this acts as token-level stochastic regularization—akin to dropout or label smoothing—improving robustness to background shifts without adding inference overhead.

B Dataset Details

B.1 Focura Dataset

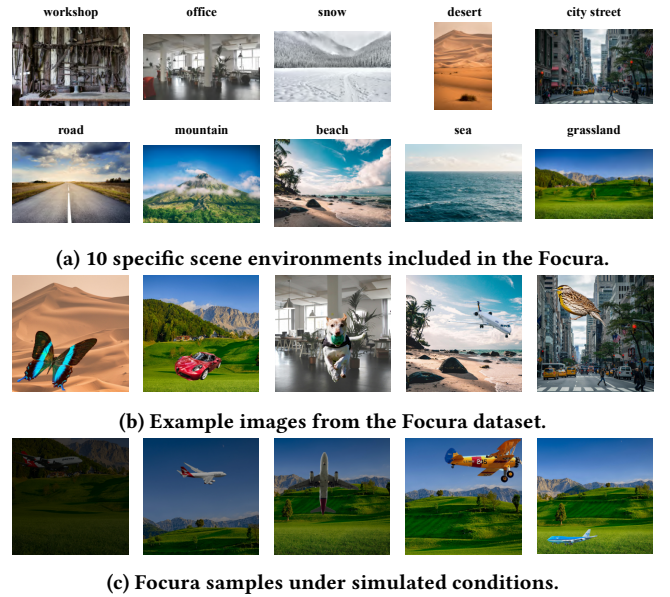


Figure 5: Illustrations of the Focura dataset.

B.1.1 Dataset Construction. We utilized the free image library Pexels to obtain the required images and applied the Segment Anything Model [21] to extract foreground objects. By combining various background scenes with different foreground objects, we created Focura. It contains 10 scenes, 10 categories, and 20 images per category to facilitate a proper evaluation of Glasses. The dataset and the loading code are publicly available on <https://github.com/ss-Zhong/Focura>. The Python scripts used to build the dataset are also available in the repository to enable the reproduction of our experimental results and to facilitate the extension of the Focura in future research.

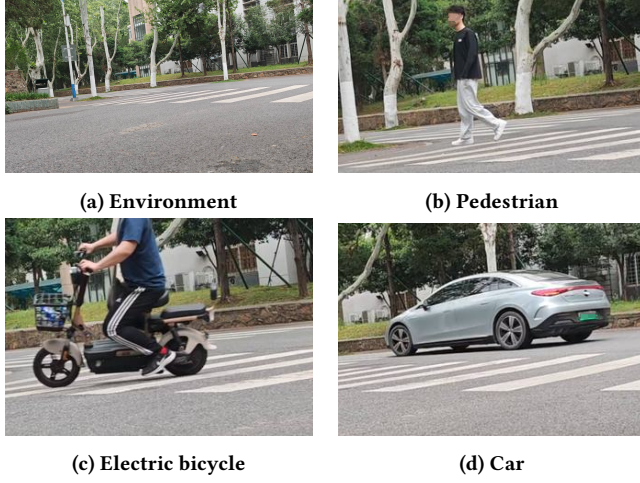


Figure 6: Example images from the Street dataset, including the overall scene environment and representative object samples.

Figure 5a illustrates the 10 scenes included in the Focura dataset, while Figure 5b presents 5 randomly selected sample images from Focura.

B.1.2 Simulation of Realistic Conditions. To approximate real-world deployment environments, we applied the following perturbations to the Focura dataset:

- **Illumination variation.** We adjusted the brightness of images using the ImageEnhance.Brightness function from the PIL library, varying the illumination factor within the range $[0.5, 1.0]$. This emulates diverse lighting conditions, such as day-to-night transitions or shadows. We further evaluated the robustness of our method under this perturbation using the ViT-S backbone.
- **Background shifts.** Each background was constructed by randomly cropping different regions from a single environment image. This simulates natural displacements caused by slight camera movements or object shifts. Although only one environment image is used for adaptation, the cropped regions yield varying visual contexts, thereby reflecting realistic background changes.

These perturbations enhance the realism of the dataset and promote model robustness to environmental variations, as shown in Figure 5c.

B.2 Street Dataset

To mitigate potential biases introduced by synthetic data and ensure the reliability of the evaluation, we constructed a real-world dataset that contains images captured in the street. This dataset consists of 5 object categories: pedestrian, bicycle, electric bicycle, electric motorcycle, and car. Due to privacy concerns involving individuals in the photos, only a few example images are presented here in Figure 6 instead of the full dataset.

C Sensitivity Analysis on the Infusion Ratio p

To substantiate the analysis in Appendix A and select a suitable infusion ratio, we conduct a single-parameter sensitivity analysis of p on ViT-S/16 using Focura; the results are reported in Table 7. The best performance is achieved at $p = 0.5$ (85.33% for 1-shot; 93.84% for 5-shot). Moreover, results remain tight across $p \in [0.3, 0.9]$, indicating low sensitivity to p and strong robustness. We therefore adopt $p = 0.5$ in the main experiments.

Table 7: Effect of the infusion ratio p on ViT-S/16. χ denote without Glasses

Setting	Infusion ratio p					
	χ	0.1	0.3	0.5	0.7	0.9
1-shot	65.85	77.20	83.35	85.33	84.88	83.26
5-shot	85.90	91.12	92.90	93.84	93.70	92.73

D Experiment Results with More Details

D.1 Performance across Different Backbones

Table 8 summarizes the scene-wise performance Glasses with different backbones on the Focura dataset. It can be observed that Glasses consistently achieves an outstanding performance across all scenes.

D.2 Details of Performance Across Different Baselines

Table 9 summarizes the scene-wise performance of different baselines on the Focura dataset.

Table 8: Accuracies of different ViT-based backbones across various Focura scenes. Average accuracies with 95% confidence intervals are reported.

Backbone	Metrics	Classes	workshop		office		snow		desert		city street		road		mountain		beach		sea		grassland	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
DeiT-T/16	$\ \cdot\ ^2$	$\times$	49.97	73.27	55.64	78.93	65.04	88.87	67.53	89.03	43.56	70.29	64.05	87.36	59.33	82.99	59.85	85.00	67.61	86.81	59.76	83.08
		$\checkmark$	± 1.99	± 1.45	± 1.95	± 1.31	± 2.17	± 1.18	± 1.81	± 0.99	± 2.02	± 1.44	± 1.76	± 1.04	± 2.52	± 1.32	± 2.12	± 1.02	± 1.78	± 1.17	± 2.08	± 1.07
			68.05	82.91	67.15	84.08	78.32	91.27	82.28	92.69	63.49	82.08	79.20	90.96	73.17	88.61	73.91	88.00	79.92	91.48	71.99	88.11
	cos	$\times$	± 1.95	± 1.32	± 2.15	± 1.24	± 1.84	± 0.90	± 1.56	± 0.73	± 2.07	± 1.12	± 1.62	± 0.79	± 2.07	± 0.82	± 1.79	± 0.87	± 1.62	± 0.83	± 2.16	± 0.79
		$\checkmark$	50.36	71.17	56.49	77.07	68.99	87.31	70.13	86.97	46.95	69.61	67.57	85.40	63.72	81.96	61.91	83.13	69.17	85.49	63.04	81.57
			± 1.93	± 1.47	± 2.03	± 1.32	± 2.19	± 1.30	± 1.71	± 1.08	± 2.02	± 1.51	± 1.62	± 1.21	± 2.25	± 1.36	± 2.19	± 1.12	± 1.86	± 1.26	± 2.06	± 1.11
DeiT-B/16	$\ \cdot\ ^2$	$\times$	50.44	78.97	56.13	84.83	63.36	92.57	63.17	91.87	45.95	80.27	62.60	92.51	57.84	88.21	61.73	89.60	64.77	91.24	62.47	92.73
		$\checkmark$	± 1.81	± 1.45	± 2.44	± 1.25	± 2.45	± 0.68	± 2.38	± 0.80	± 2.16	± 1.48	± 2.38	± 0.76	± 2.58	± 0.98	± 2.21	± 0.84	± 2.39	± 0.83	± 2.54	± 0.74
			77.73	90.08	76.56	91.59	82.83	95.21	87.59	97.57	72.33	90.13	84.69	95.85	82.17	94.55	80.79	94.28	85.68	95.39	80.45	94.88
	cos	$\times$	± 1.81	± 0.93	± 2.06	± 0.85	± 1.46	± 0.60	± 1.31	± 0.40	± 2.24	± 0.95	± 1.36	± 0.54	± 1.68	± 0.59	± 1.91	± 0.65	± 1.47	± 0.53	± 1.78	± 0.60
		$\checkmark$	50.53	77.45	61.77	83.23	74.36	91.00	71.45	91.29	54.77	80.00	72.56	90.72	68.93	88.48	67.33	88.80	72.33	89.43	72.93	91.03
			± 1.76	± 1.55	± 2.22	± 1.28	± 1.78	± 0.99	± 1.66	± 0.81	± 2.16	± 1.40	± 1.65	± 0.89	± 1.99	± 0.98	± 2.04	± 0.80	± 1.92	± 1.03	± 1.90	± 0.87
ViT-B/16	$\ \cdot\ ^2$	$\times$	54.71	76.53	66.41	84.55	81.83	93.81	81.19	93.16	56.59	78.09	78.05	93.11	66.27	87.20	77.12	90.68	81.51	93.04	75.04	92.31
		$\checkmark$	± 2.22	± 1.54	± 2.15	± 1.37	± 1.72	± 0.66	± 1.77	± 0.79	± 2.18	± 1.37	± 1.67	± 0.65	± 2.12	± 1.05	± 1.74	± 0.69	± 1.80	± 0.79	± 2.23	± 0.95
			71.08	85.53	79.79	91.67	81.83	93.97	89.39	96.84	70.91	87.80	85.65	95.31	82.80	93.76	82.29	93.45	87.25	95.49	81.25	93.88
	cos	$\times$	± 2.07	± 1.18	± 1.69	± 0.91	± 1.76	± 0.62	± 1.22	± 0.40	± 2.11	± 1.04	± 1.57	± 0.53	± 1.42	± 0.62	± 1.82	± 0.63	± 1.50	± 0.49	± 1.94	± 0.65
		$\checkmark$	54.35	75.09	66.72	83.36	82.07	93.51	80.76	93.44	56.80	77.19	77.76	92.04	66.92	86.79	76.96	90.84	81.33	92.80	74.83	92.12
			± 2.19	± 1.60	± 2.11	± 1.36	± 1.72	± 0.67	± 1.80	± 0.75	± 2.21	± 1.39	± 1.70	± 0.71	± 2.06	± 1.06	± 1.68	± 0.78	± 1.73	± 0.75	± 2.19	± 0.93
Swin-T/7	$\ \cdot\ ^2$	$\times$	54.83	82.16	62.59	87.28	71.25	93.81	62.32	91.89	50.15	80.71	67.52	92.01	69.41	92.01	64.60	90.63	71.53	92.65	69.53	93.19
		$\checkmark$	± 1.94	± 1.50	± 2.11	± 1.11	± 2.17	± 0.67	± 1.98	± 0.94	± 2.08	± 1.54	± 1.96	± 0.99	± 2.41	± 0.81	± 2.11	± 0.77	± 2.02	± 0.81	± 2.20	± 0.82
			72.84	89.64	81.17	93.12	84.31	95.48	88.24	97.35	69.71	89.09	84.63	96.07	85.95	95.47	83.47	95.91	87.15	97.03	85.12	96.41
	cos	$\times$	± 2.02	± 1.09	± 1.82	± 0.67	± 1.83	± 0.54	± 1.37	± 0.42	± 2.29	± 1.04	± 1.74	± 0.46	± 1.73	± 0.44	± 1.88	± 0.54	± 1.56	± 0.37	± 1.63	± 0.50
		$\checkmark$	58.93	81.85	65.77	85.49	79.00	92.84	68.01	91.21	55.16	79.37	73.85	91.52	75.77	92.40	70.81	91.25	77.64	92.51	73.52	91.69
			± 2.01	± 1.54	± 2.03	± 1.26	± 1.86	± 0.89	± 1.92	± 0.96	± 1.99	± 1.47	± 1.76	± 0.90	± 1.88	± 0.80	± 2.23	± 0.86	± 1.89	± 0.80	± 1.87	± 1.02
Swin-S/7	$\ \cdot\ ^2$	$\times$	51.94	88.91	82.92	92.44	87.27	95.12	90.49	97.11	73.96	88.60	88.84	95.48	88.36	95.33	86.48	95.55	90.13	96.92	87.29	95.85
		$\checkmark$	± 1.84	± 1.07	± 1.63	± 0.72	± 1.53	± 0.56	± 1.22	± 0.43	± 2.06	± 1.01	± 1.36	± 0.53	± 1.35	± 0.45	± 1.53	± 0.60	± 1.27	± 0.39	± 1.46	± 0.50
			49.44	79.09	61.20	88.23	68.48	92.92	63.75	91.09	47.91	79.41	65.48	89.44	64.37	88.49	64.85	90.43	71.68	92.45	67.15	91.89
	cos	$\times$	± 1.98	± 1.63	± 2.14	± 0.93	± 2.33	± 0.68	± 2.01	± 0.95	± 2.06	± 1.35	± 1.87	± 0.99	± 2.15	± 1.03	± 2.21	± 0.85	± 2.03	± 0.82	± 2.06	± 0.89
		$\checkmark$	77.79	90.21	81.87	93.41	87.32	96.39	90.57	97.71	75.11	91.61	87.79	96.19	86.51	96.28	86.64	95.93	90.67	97.64	85.69	96.43
			± 1.94	± 1.19	± 1.78	± 0.71	± 1.66	± 0.46	± 1.47	± 0.36	± 1.88	± 0.91	± 1.54	± 0.44	± 1.69	± 0.49	± 1.60	± 0.49	± 1.29	± 0.36	± 1.57	± 0.50
Swin-T/7	$\ \cdot\ ^2$	$\times$	52.17	79.16	67.77	87.63	77.45	92.75	71.59	92.61	56.01	80.81	73.53	90.92	71.80	88.79	73.96	90.71	77.80	92.08	72.53	91.49
		$\checkmark$	± 2.05	± 1.59	± 2.01	± 1.02	± 1.81	± 0.88	± 1.92	± 0.77	± 2.11	± 1.21	± 1.94	± 0.87	± 1.89	± 0.98	± 2.07	± 0.82	± 2.04	± 0.93	± 2.18	± 1.01
			78.45	89.48	83.41	92.83	89.52	95.95	93.00	97.83	77.27	90.92	89.77	95.92	89.20	96.25	89.40	95.84	92.95	97.60	88.19	96.01
	cos	$\times$	± 1.87	± 1.17	± 1.58	± 0.75	± 1.32	± 0.46	± 1.09	± 0.35	± 1.73	± 0.94	± 1.33	± 0.43	± 1.32	± 0.47	± 1.22	± 0.51	± 1.11	± 0.37	± 1.33	± 0.54
		$\checkmark$																				

Table 9: Accuracies of different baselines across various Focura scenes. Average accuracies with 95% confidence intervals are reported.

Model	Backbone	Pre-training Task	workshop		office		snow		desert		city street		road		mountain		beach		sea		grassland	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ProtoNet _f	ViT-S/16	classification	50.44	78.97	56.13	84.83	63.36	92.57	63.17	91.87	45.95	80.27	62.60	92.51	57.84	88.21	61.73	89.60	64.77	91.24	62.47	92.73
ProtoNet _{cos}	ViT-S/16	classification	±1.81	±1.45	±2.44	±1.25	±2.45	±0.68	±2.38	±0.80	±2.16	±1.48	±2.38	±0.76	±2.58	±0.98	±2.21	±0.84	±2.39	±0.83	±2.54	±1.04
			50.53	77.45	61.77	83.23	74.36	91.00	71.45	91.29	54.77	80.00	72.56	90.72	68.93	88.48	67.33	88.80	72.33	89.43	72.93	91.03
FewTURE	ViT-S/16	iBOT	±1.76	±1.55	±2.22	±1.28	±1.78	±0.99	±1.66	±0.81	±2.16	±1.40	±1.65	±0.89	±1.99	±0.98	±2.04	±0.90	±1.92	±1.03	±1.90	±0.87
			36.05	50.19	32.09	44.41	65.76	63.90	51.91	64.96	30.69	36.20	46.09	63.87	48.77	70.04	43.31	61.27	51.05	69.44	47.88	61.77
CPEA	ViT-S/16	iBOT	±1.49	±1.43	±1.44	±1.54	±1.59	±1.42	±1.71	±1.48	±1.05	±1.27	±1.81	±1.49	±1.65	±1.49	±1.72	±1.58	±1.44	±1.64	±1.56	±1.46
			65.40	83.93	69.27	86.20	77.92	92.64	79.57	93.77	62.21	83.09	76.57	92.56	76.89	92.29	74.37	91.31	76.83	91.67	73.76	90.71
BPA	ViT-S/16	iBOT	±1.23	±1.56	±1.30	±1.61	±1.46	±1.72	±1.49	±1.75	±1.17	±1.55	±1.43	±1.72	±1.44	±1.72	±1.39	±1.70	±1.44	±1.71	±1.38	±1.69
			63.71	82.11	70.63	87.53	77.20	92.65	76.19	92.40	61.28	82.27	76.48	91.39	72.23	90.36	73.36	91.21	79.21	93.36	73.56	91.48
			±2.18	±1.34	±2.08	±1.01	±1.75	±0.83	±1.76	±0.83	±2.15	±1.12	±1.76	±0.83	±2.02	±0.82	±1.73	±0.81	±1.77	±0.78	±1.94	±0.88
 ProtoNet _{cos}	ViT-S/16	classification	79.89	89.67	80.29	91.41	84.76	94.76	90.28	97.51	76.93	89.45	86.89	95.29	84.91	94.13	85.12	94.37	88.35	94.95	84.55	94.48
 ProtoNet _{cos}	ViT-S/16	iBOT	±1.72	±0.92	±1.76	±0.80	±1.36	±0.67	±1.16	±0.42	±1.84	±0.94	±1.24	±0.56	±1.42	±0.63	±1.55	±0.58	±1.35	±0.57	±1.48	±0.59
			79.60	89.47	83.80	92.56	85.27	95.01	88.29	96.09	76.96	88.69	87.33	95.19	85.40	95.00	84.91	94.04	88.08	95.32	84.20	94.41
			±1.77	±1.04	±1.70	±0.68	±1.37	±0.64	±1.42	±0.66	±2.01	±1.00	±1.36	±0.62	±1.38	±0.59	±1.39	±0.69	±1.26	±0.64	±1.47	±0.70